DSCI 510 – Final Project Progress Report
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Can Qualifying Position Predict Race Results in Recent Formula 1 Seasons?

Project Scope Update

The core research question remains the same: whether qualifying position can predict Formula 1 race finishing position during the modern ground-effect regulation era. The current implementation focuses on the 2023–2025 seasons because OpenF1 historical coverage begins in 2023.

The project has expanded from a single-source qualifying analysis into a multi-source predictive modeling project. In addition to qualifying position, the final model incorporates pit stop counts, rainfall, track temperature, and circuit type to evaluate whether race-context variables improve prediction performance.

The completed pipeline includes multi-source data collection, preprocessing, feature engineering, exploratory analysis, regression modeling, diagnostics, and combined-model comparison. The project also includes a simple strategy-simulation extension for hypothetical race scenarios.

Data Sources

The project combines three Formula 1 data sources:

- OpenF1 API (https://api.openf1.org/v1): Used for qualifying position, finishing position, driver information, and pit stop counts.
- FastF1 Python Library (https://docs.fastf1.dev): Used for weather-related variables such as rainfall and track temperature.
- Kaggle (https://www.kaggle.com/datasets/rokyt07/formula-1-championships-1950-2025): Used for supplementary circuit metadata and circuit-type classification.

After preprocessing and merging, the final dataset contains approximately 1,200 driver-race observations from the 2023–2025 seasons.

Current Results

The analysis confirms that qualifying position is a strong predictor of race finishing position. The baseline qualifying-only model achieved a Pearson correlation of approximately 0.75 and an R^2 value around 0.56. Front-grid starters also showed substantially higher win rates and better average finishing positions.

Additional features such as pit stops, weather, and circuit type showed weaker standalone predictive power. The final combined model slightly improved prediction performance, but qualifying position remained the dominant predictor in feature-importance analysis.

Issues / Difficulties

The main challenge was integrating multiple data sources with inconsistent race naming conventions and incomplete endpoint coverage. Some OpenF1 pit-stop requests returned missing sessions or 404 responses, requiring additional error handling and fallback logic.

Another challenge was balancing interpretability and predictive performance. While linear regression provides clear interpretation, Formula 1 race outcomes still contain substantial randomness caused by incidents, safety cars, weather changes, and race strategy variability.